Biochemistry Cume April 17, 2009

The questions below are based upon a recent departmental seminar on the topic of lipoic acid biosynthesis.

1. (30 points) Draw structures for oxidized and reduced forms of lipoic acid.
2. (10 points) When bound to protein, lipoic acid is covalently attached to the side chain of which amino acid?
3. (15 points) Draw a covalent adduct between lipoic acid and the side chain that is relevant for question two, above.
4. (15 points) It was mentioned that a particular fatty acid is a precursor of lipoic acid during its biosynthesis in E. coli. Name and draw this fatty acid.

   (Hint: Both molecules have the same number of carbon atoms.)
5. (10 points) Describe a role for methionine during the biosynthesis of lipoic acid.
6. (10 points) Describe a role for iron during the biosynthesis of lipoic acid.
7. (10 points) Based upon the seminar or other information, define the term "prochiral."